WORKED SOLUTION

STEP I 2005, Question 10

Mechanics • Collisions and Restitution

STEP I 2005, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

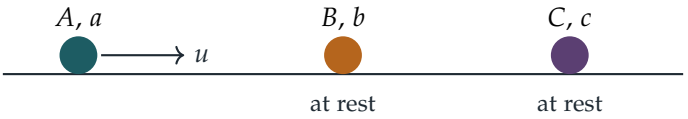
step.maths.org/questions/05-s1-q10

The key idea

Each collision is a direct impact between two smooth particles, so it gives us two equations: conservation of momentum along the line, and Newton's experimental law (separation speed $= e\times$ approach speed). Solving these once for a general pair gives formulas we can reuse for the second collision. For the special cases, the real question is whether the story ends after two collisions: we need the final velocities, so we must check that no particle catches up with the one in front.

Solution

We model A, B, C as particles moving along one line on a smooth surface, so no horizontal external forces act and momentum along the line is conserved in each impact. Take the direction of u as positive. Write p for the velocity of A after its collision with B, and q for the velocity of B after its collision with C.



The first collision: finding v

Before impact A has velocity u and B is at rest; afterwards they have velocities p and v. Momentum and Newton's law give

$$au = ap + bv, \qquad v - p = eu.$$

Substituting $p = v - eu$ into the first equation, $au = av - aeu + bv$, so

$$v = \frac{a(1+e)u}{a+b}, \qquad p = v - eu = \frac{(a-eb)u}{a+b}.$$

Note $v > 0$, so B really does move towards C.

The second collision: finding w

This is the same situation with B (mass b, speed v) striking C (mass c, at rest). So we can reuse the formulas above with $a \to b$, $b \to c$, $u \to v$:

$$w = \frac{b(1+e)v}{b+c}, \qquad q = \frac{(b-ec)v}{b+c}.$$

Substituting for v,

$$w = \frac{b(1+e)}{b+c} \cdot \frac{a(1+e)u}{a+b} = \frac{abu(1+e)^2}{(a+b)(b+c)},$$

as required.

After these two collisions the velocities are p, q, w. There is a further collision only if $p > q$ (A catches B) or $q > w$ (B catches C); since $q - w = -ev < 0$ by Newton's law, the second never happens, so we only need to compare p and q.

Ask yourself Why can't we just stop after two collisions and quote p, q, w?

Because the question asks for *final* velocities. If A were still moving faster than B afterwards, it would hit B again and everything would change. So in each case we must check that $p \le q$.

Part (i)

Here $a = eb$ and $b = ec$ (with $e > 0$, as the masses are positive). Then $a - eb = 0$ and $b - ec = 0$, so

$$p = 0, \qquad q = 0.$$

For w, use $a + b = b(1+e)$ and $b + c = b + b/e = b(1+e)/e$:

$$w = \frac{eb \cdot b\,u(1+e)^2}{b(1+e) \cdot b(1+e)/e} = e^2 u.$$

A and B are both at rest and C moves away, so there are no more collisions. The final velocities are

$$A:\ 0, \qquad B:\ 0, \qquad C:\ e^2 u.$$

Each collision passes all the momentum on to the next particle, like a Newton's cradle with lighter particles behind.

Part (ii)

Now $b = ea$ and $c = eb = e^2 a$. Then

$$v = \frac{a(1+e)u}{a(1+e)} = u, \qquad p = \frac{(a - e^2 a)u}{a(1+e)} = (1-e)u,$$

using $1 - e^2 = (1-e)(1+e)$. In the second collision the masses b, c are in the same ratio as a, b and B arrives with speed u, so it is an exact copy of the first collision:

$$w = u, \qquad q = (1-e)u.$$

(Directly: $w = \dfrac{a \cdot ea \cdot u(1+e)^2}{a(1+e) \cdot ea(1+e)} = u$.)

Now $p = q$, so A never catches B; and $q = (1-e)u < u = w$ for $e > 0$, so B never catches C. The final velocities are

$$A:\ (1-e)u, \qquad B:\ (1-e)u, \qquad C:\ u,$$

all in the original direction of motion of A.

Checks

- Momentum in (ii): $a(1-e)u + ea(1-e)u + e^2 a\, u = au$. In (i): $c \cdot e^2 u = e^2 c\, u = au$.
- With $e = 1$ in (ii) (perfectly elastic), A and B end at rest and C moves off with u; kinetic energy is conserved: $\frac{1}{2}au^2 = \frac{1}{2}cu^2$ needs $c = a$, which is exactly $c = e^2 a$ with $e = 1$.
- A numerical case: with $e = \frac{1}{2}$, masses $1, 2, 4$ and $u = 1$ (part (i)) the final velocities are $0, 0, \frac{1}{4}$; masses $4, 2, 1$ (part (ii)) give $\frac{1}{2}, \frac{1}{2}, 1$. We confirmed these, and the general formulas for v and w, with exact arithmetic over thousands of random masses and values of e, simulating any further impacts.

Common slips

- Writing Newton's law with the velocities the wrong way round: it is (speed of the front particle − speed of the back particle) after $= e \times$ (back − front) before.
- Quoting the velocities after two collisions without checking whether A catches B again.

Beyond STEP

Chains of collisions are the simplest model of how momentum and energy travel through a material, and the mass ratios in part (i) are chosen so that *all* the momentum ends up in the last particle. Ordering the masses the other way, heavy behind light, with e close to 1, makes the last particle move faster than the first: the idea behind the toy where a small ball balanced on a large one shoots high into the air when the pair is dropped. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats collisions using the centre-of-mass frame, which makes formulas like v and w almost obvious.

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